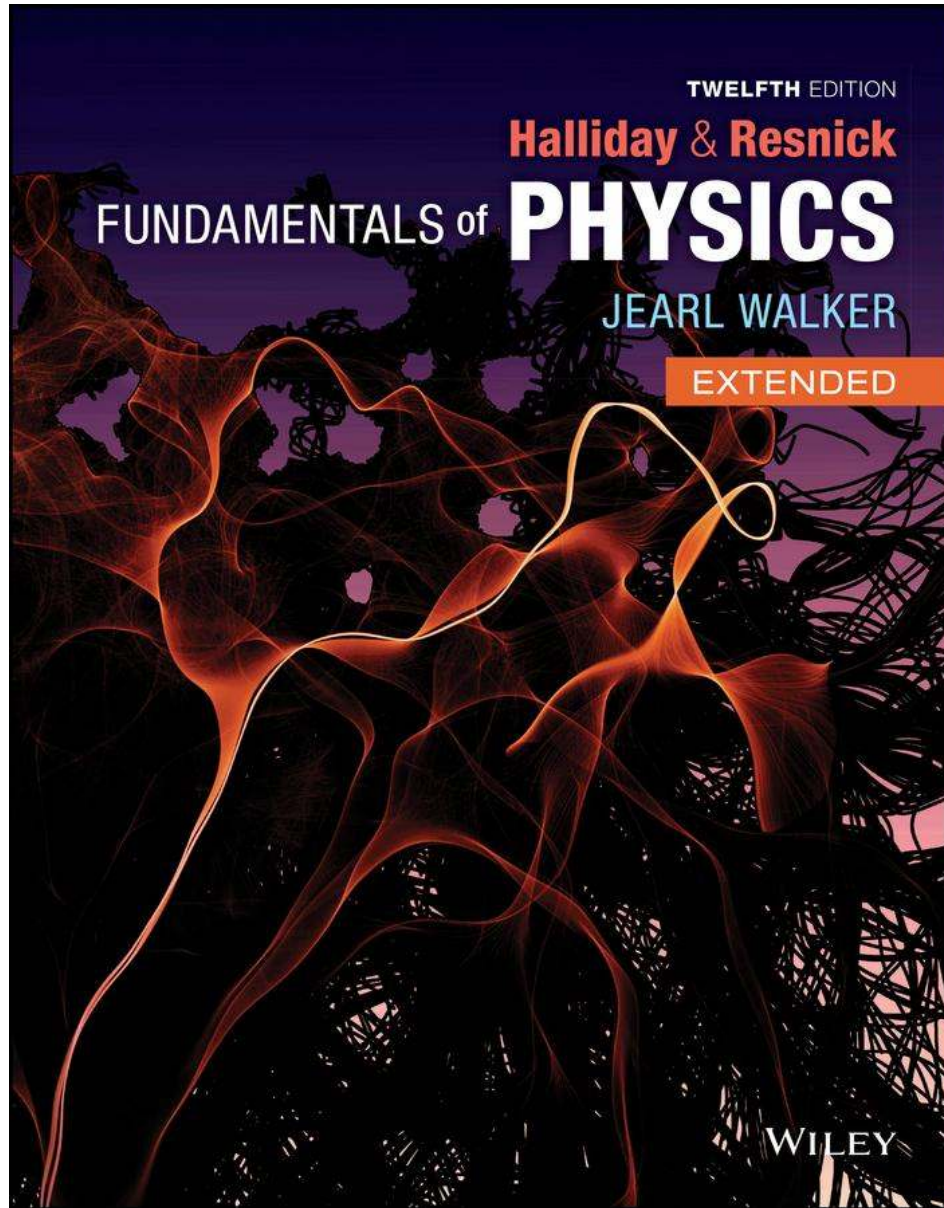




Fundamentals of Physics

Twelfth Edition

Halliday & Resnick

Jearl Walker

Chapter 21

Coulomb's Law

Units of Chapter 21

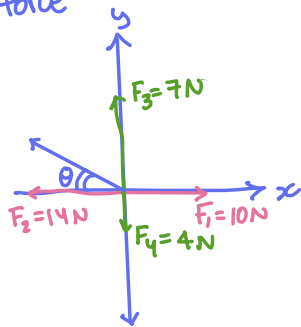
21.1 Coulomb's Law

21.2 Charge is Quantized

21.3 Charge is Conserved

Warm up:
Find the net force

polar
coordinates



$$\sum F_x = 10 + (-14) = -4 \text{ N left or } -\hat{x}$$

$$\sum F_y = 7 - 4 = 3 \text{ N upward or } +\hat{y}$$

$$\text{Magnitude: } \sqrt{(-4)^2 + 3^2} = 5 \text{ N}$$

$$\text{Direction: } \tan^{-1}\left(\frac{3}{4}\right) = -36.8^\circ$$

Students mistake:

- Never memorize

overthinking
is the enemy

F_{net}/Total

① Forces in 1D (x or y-axis)



• only x or only y
+ Same
- opp
has lots of grade

② Forces are perpendicular



$$\text{magnitude} \rightarrow F_{\text{net}} = \sqrt{3^2 + 4^2} = 5 \text{ N}$$

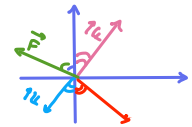
$$\text{direction} \rightarrow \tan^{-1}\left(\frac{F_y}{F_x}\right): \tan^{-1}\left(\frac{4}{3}\right) = \theta$$

Section 21.1 Coulomb's Law

Learning Objectives

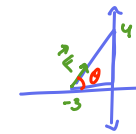
1. Distinguish between being electrically neutral, negatively charged, and positively charged.
2. Distinguish between conductors, nonconductors (insulators), semiconductors, and superconductors.
3. Identify that Coulomb's law applies only to (point-like) particles and objects that can be treated as particles.
4. If more than one force acts on a particle, find the net force by adding all the forces as vectors, not scalars.

(mag, θ)
from +ve x-axis
counterclockwise



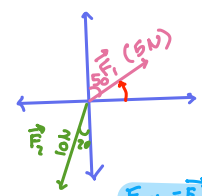
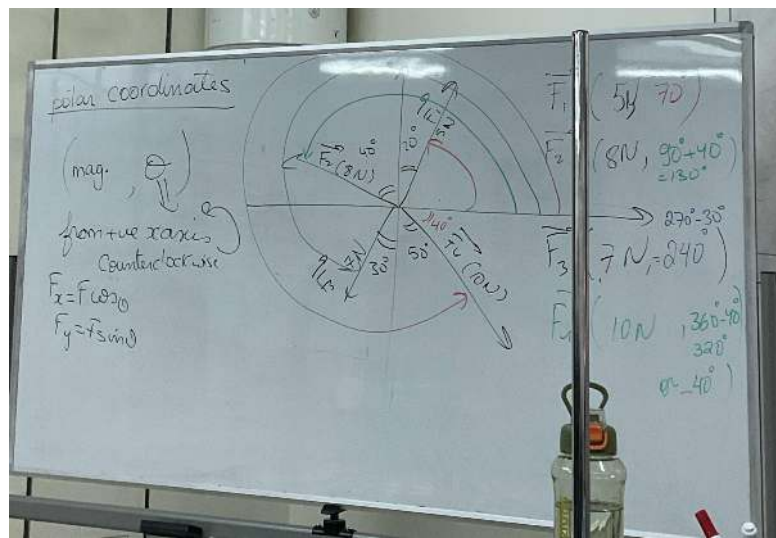
$$F_x = F \cos \theta$$

$$F_y = F \sin \theta$$



$\theta = \tan^{-1}\left(\frac{4}{-3}\right)$ if they are both F_{netx}
 F_{nety}
 $\theta = \tan^{-1}\left(\frac{4}{-3}\right)$ because it's a regular triangle
only hyp is the vector
rest is dimensions

angle θ not given



$$\vec{F}_1 (5, 40^\circ) \begin{cases} F_{1x} = F_1 \cos \theta = 5 \cos(40) = +3.8 \text{ N } (+\hat{x}) \\ F_{1y} = F_1 \sin \theta = 5 \sin(40) = +3.2 \text{ N } (+\hat{y}) \end{cases}$$

$$\vec{F}_2 (10, 250^\circ) \begin{cases} F_{2x} = 10 \cos(250) = -3.4 \text{ N } (-\hat{x}) \\ F_{2y} = 10 \sin(250) = -9.3 \text{ N } (-\hat{y}) \end{cases}$$

$$F_{netx} = F_{1x} + F_{2x}$$

$$+3.8 - 3.4 = +0.4 \text{ N } (+\hat{x})$$

$$F_{nety} = F_{1y} + F_{2y} = 3.2 - 9.3 = -6.1 \text{ N } (-\hat{y})$$

$$F_{net} = \sqrt{F_{netx}^2 + F_{nety}^2}$$

$$= \sqrt{0.4^2 + (-6.1)^2}$$

$$\theta = \tan^{-1}\left(\frac{F_{nety}}{F_{netx}}\right) = \tan^{-1}\left(\frac{-6.1}{0.4}\right) = 9^\circ$$

positive $\rightarrow$ keep
negative $\rightarrow$ add 360°

BUT not imp

if x -ve $+180$

normally

polarization: redistribution of charges
+ take 1 pole
- take other pole
opposite attract
same repel
attraction is more than repulsion
repulsion weaker

atoms
Never lose or gain protons
atoms are naturally neutral
same + & -
repel
attract

rod
gained e^- -vely charged
lost e^- +vely charged
fur

rubbing becomes

Total before rub: 10
Total after rub: 10

conservation of charge

An uncharged amber rod ...

... has no effect on bits of paper.

(a)

We transfer charge to the rod by rubbing it on fur.

(b)

The charged rod ...
... now attracts the paper.

(c)

- We are all made up of atoms, and every atom in the human body contains both positive and negative electric charges.
- The effects of electric charge were first observed as static electricity ^{don't move}
- After being rubbed on a piece of fur, an amber rod acquires a charge and can attract small objects (Why?) ^{Distance}

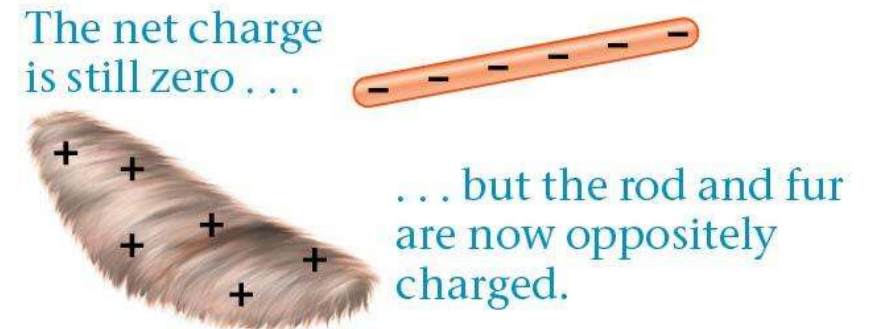
When an amber rod is rubbed with fur, some of the electrons on the atoms in the fur are transferred to the amber.



(a)



(b)



(c)

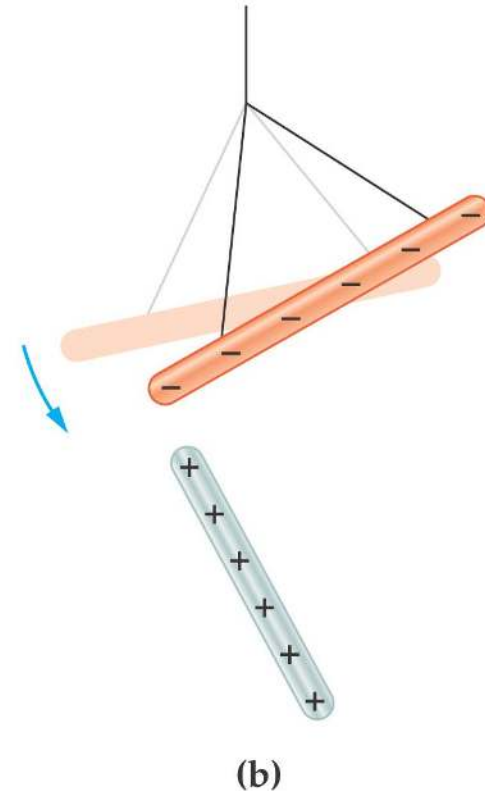
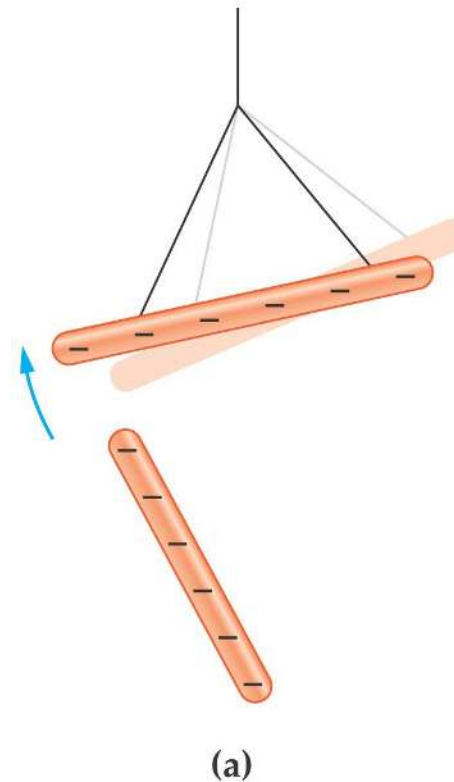
Remarks

1. The amber rod is not unique in its ability to become charged. Other materials can behave in such way as well.
2. If glass is rubbed with a piece of silk, it too can attract small objects (**the glass rod will become charged**)
3. They have noticed that, when suspending the amber and the glass rods they tend to get closer to each other, in other words, attract each other.
4. The above note implies that, the two rods (the glass and the amber) are both charged, still, **they are not totally alike**.

General properties of charged objects

1. Two rods with opposite charges will attract each other. (**Fig. a**)
2. Two rods with the same charges will repel each other. (**Fig. b**)

Question: why the amber rod became negatively charged while the glass rod became positively charged?



3. All electrons have exactly the same charge, the charge of the electron is defined as:

$$-e = -1.6 \times 10^{-19} \text{ Coulomb (C)}$$

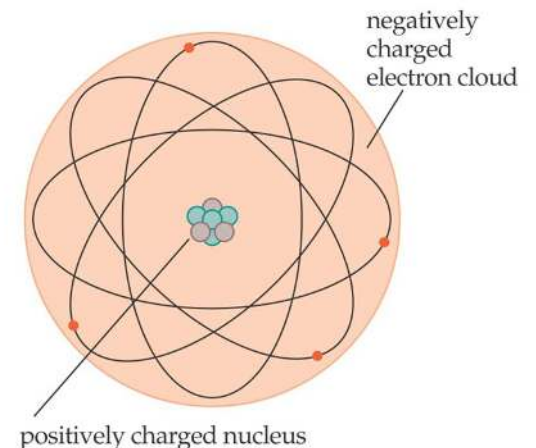
memorize

*Test in table
about units*

4. The charge of the proton (in the atomic nucleus) has the same magnitude but the opposite sign.

5. A more common unit of charge is the micro coulomb (μC) $1 \mu\text{C} = 10^{-6} \text{ C}$

6. The electrons in an atom are in a cloud surrounding the nucleus and can be separated from the atom with relative ease.



CHECKPOINT :

gain e^-
increased
↑ mass

lose e^-
decreases
↑

IMP

Is the mass of the amber rod after charging with fur:

- a) Greater than its mass before charging increase
- b) Less than its mass before charging
- c) The same as its mass before charging

Answer: a)

Note: the charged amber rod has acquired electrons from the fur, each electron has a small, but nonzero mass, and so, the mass of the amber rod increases ($m_e = 9.1 \times 10^{-31} \text{ kg}$)

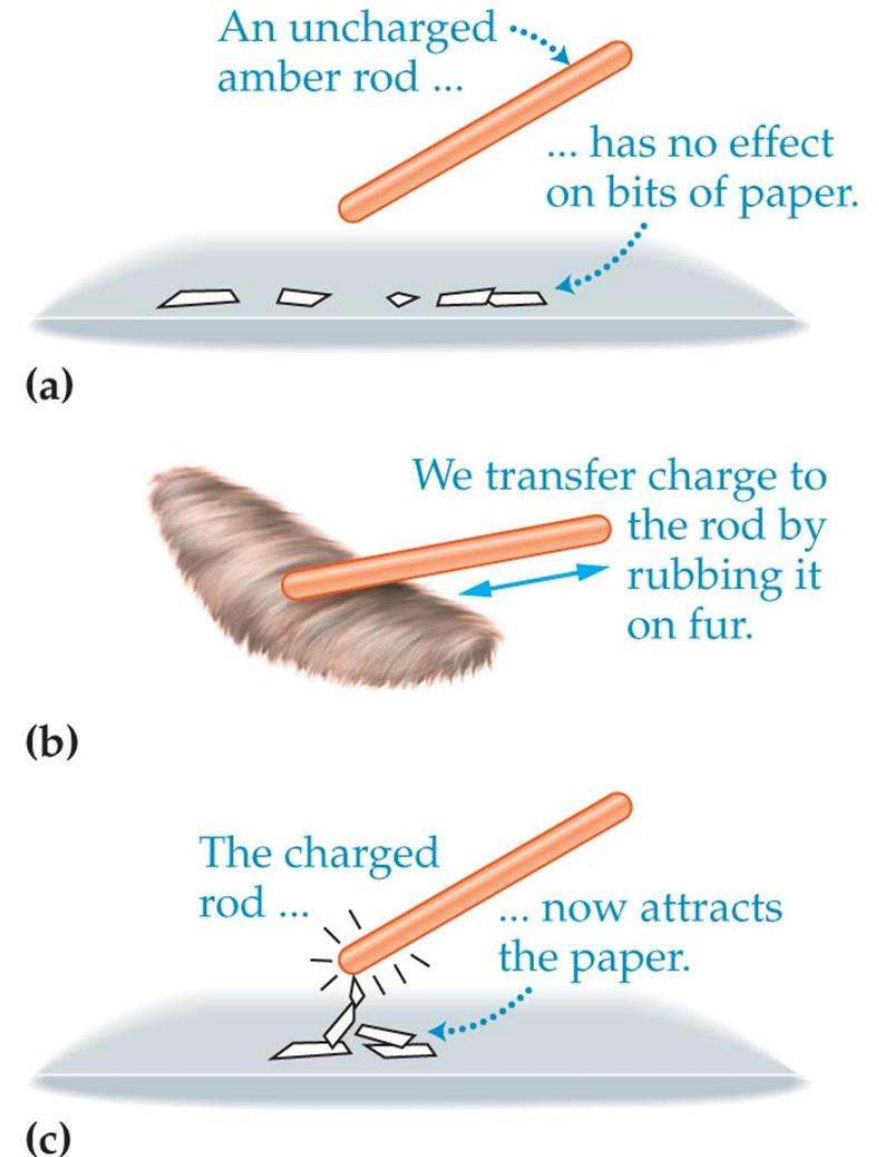
memorize?

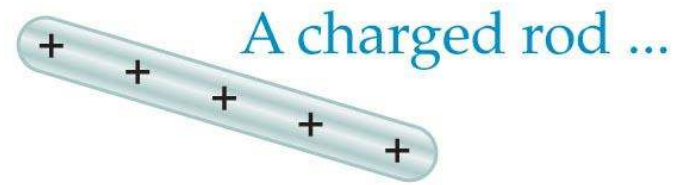
POLARIZATION (***Induced Charge***)

Question: Is it possible for a charged rod to attract small objects that have **ZERO net charge**?

Answer: yes, just like when the amber rod attracted the small pieces of paper, and the glass rod attracts small objects.

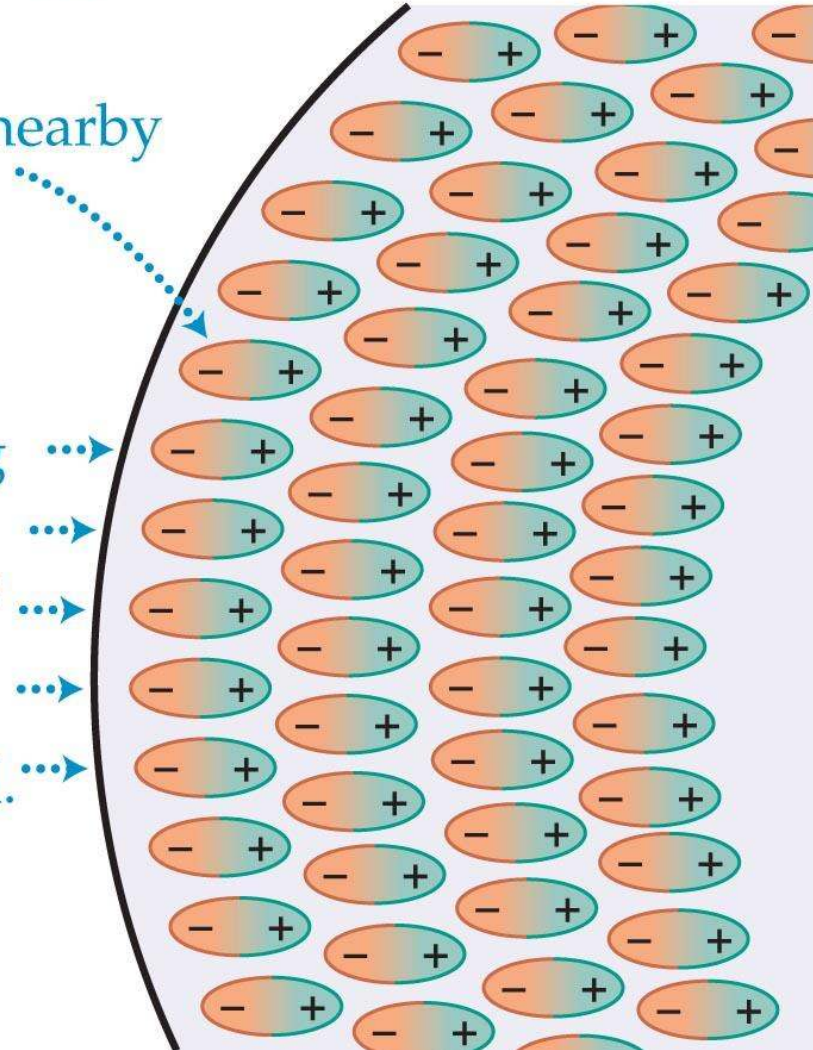
And this mechanism is called **POLARIZATION**.





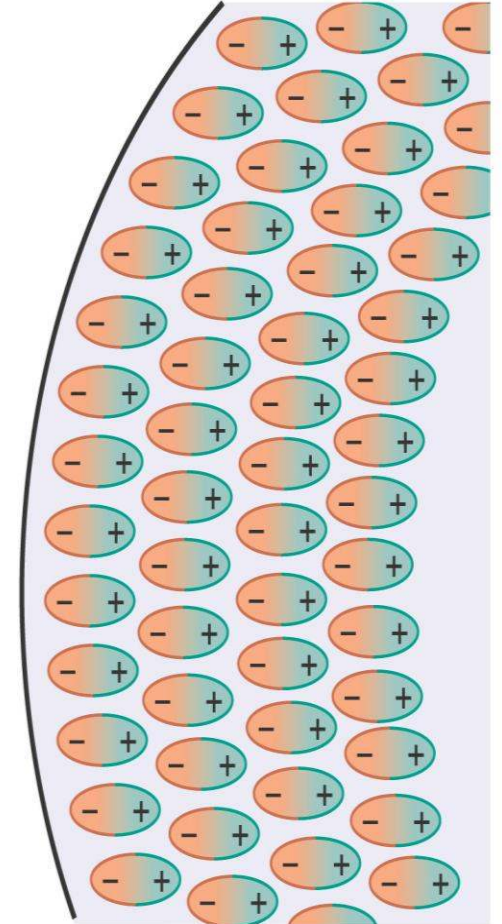
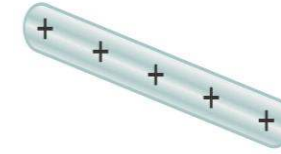
... distorts nearby
atoms ...

... producing
an excess of
the opposite
charge on
the surface
of a material.



Properties of polarized a material

1. The net charge on it is **zero**.
same e^- p^+
2. Due to the charged rod, the atoms near the surface of the polarized material will be distorted (will elongate), where, the opposite charge will rotate toward the surface, and the same charge will be repelled away from it.
3. As a result of step 2, a net opposite charge develops on the surface near the charged rod.
4. This net opposite charge is called: **induced polarized charge**



Materials classified based on their ability to move charge

1. Insulators: *e⁻ can't leave tightly bound to nucleus*

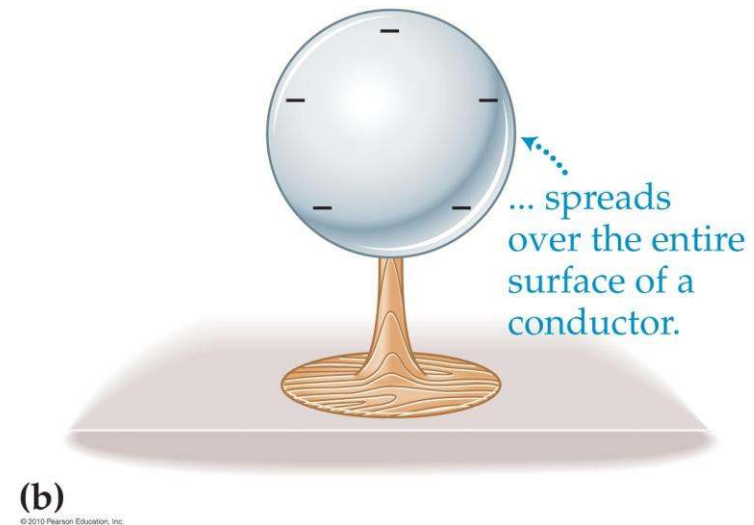
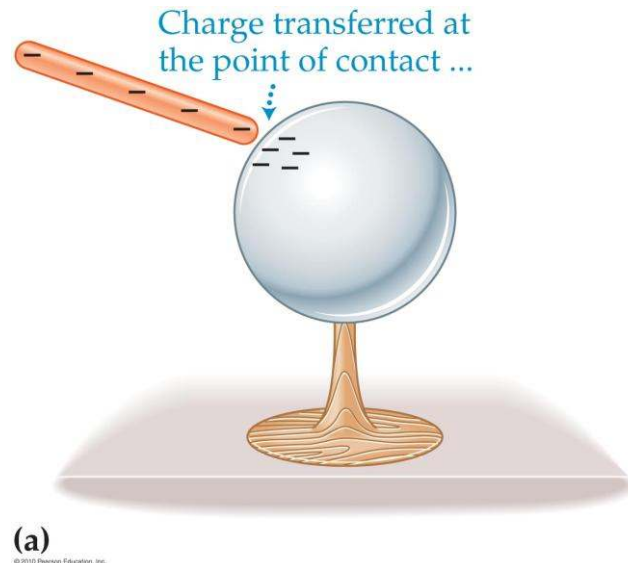
- Insulators are materials, in which charges are not free to move.
- Most insulators are nonmetallic substances.

Example: when rubbing the amber rod, the rubbed side becomes charged, whereas the other end remains neutral.

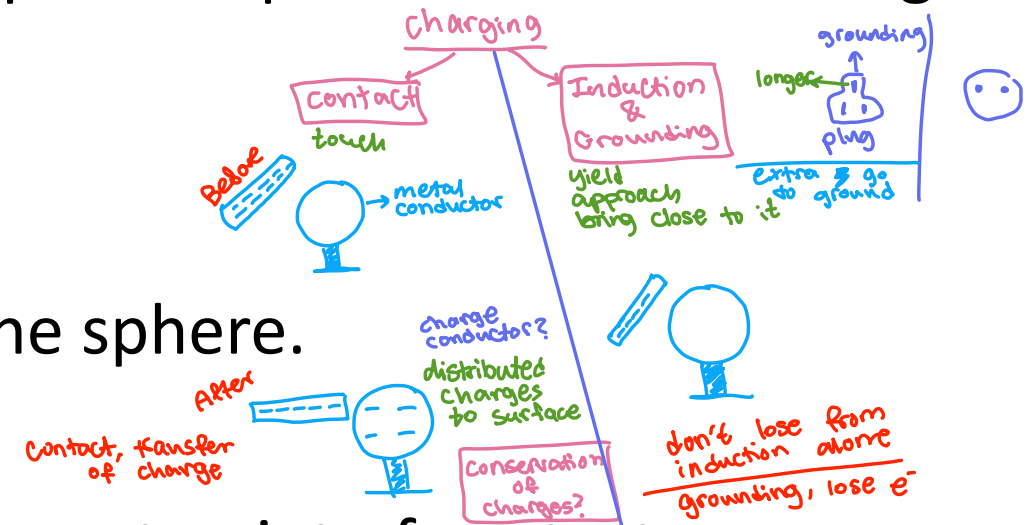
2. Conductors:

c of e^-
sea of e^-
free e^- |
move
freely

- Materials that allow the charges to move more freely. ([look at the figure](#))
- Most conductors are metals.



- In the previous figure, an uncharged metal sphere is placed in an insulating base.
- A charged rod is brought into contact with the sphere.
- Some charges will be transferred to the sphere at point of contact.
- Since the metal is a good conductor of electricity, the charges will be evenly distributed **over the whole surface of the sphere**.



3. Semiconductors

Solar cells made up semic
between insulator & conductor
can control, give energy
take away } gives electricity
to turn on things

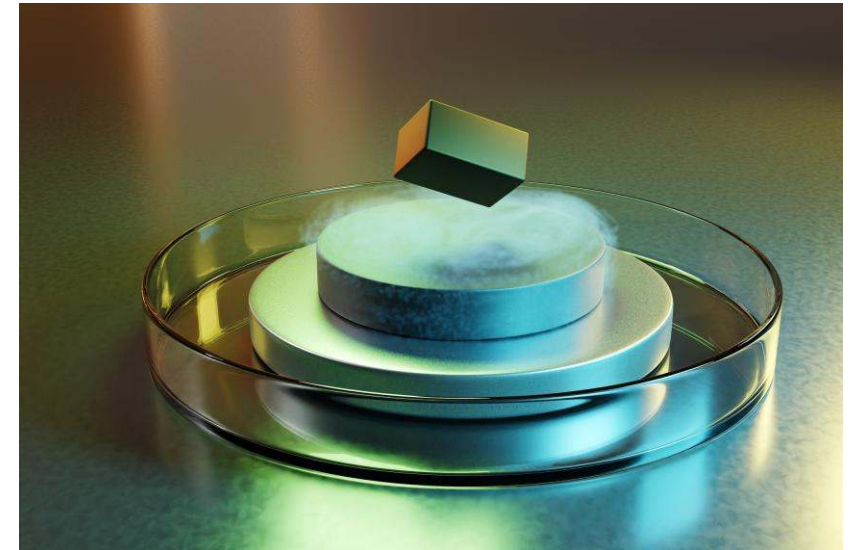
➤ Materials that are intermediate between conductors and insulators; examples include silicon and germanium in computer chips.



4. Superconductors

always 0% loss of electricity in cable
unlike conductors which have ~40%
BUT condition is low temp

➤ Materials that are perfect conductors, allowing charge to move without any hindrance



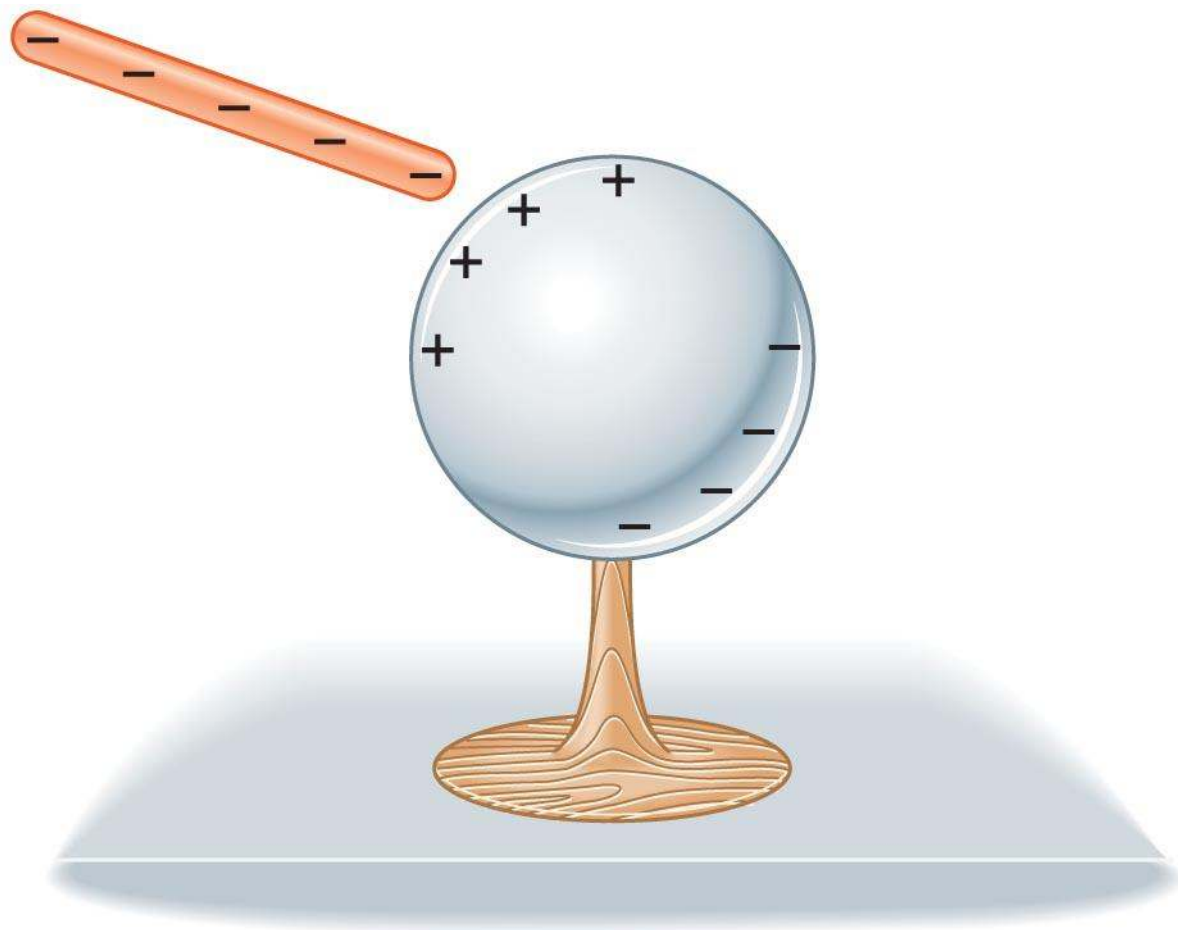
Charging By Induction

Charging by Induction is to charge an object without making direct physical contact

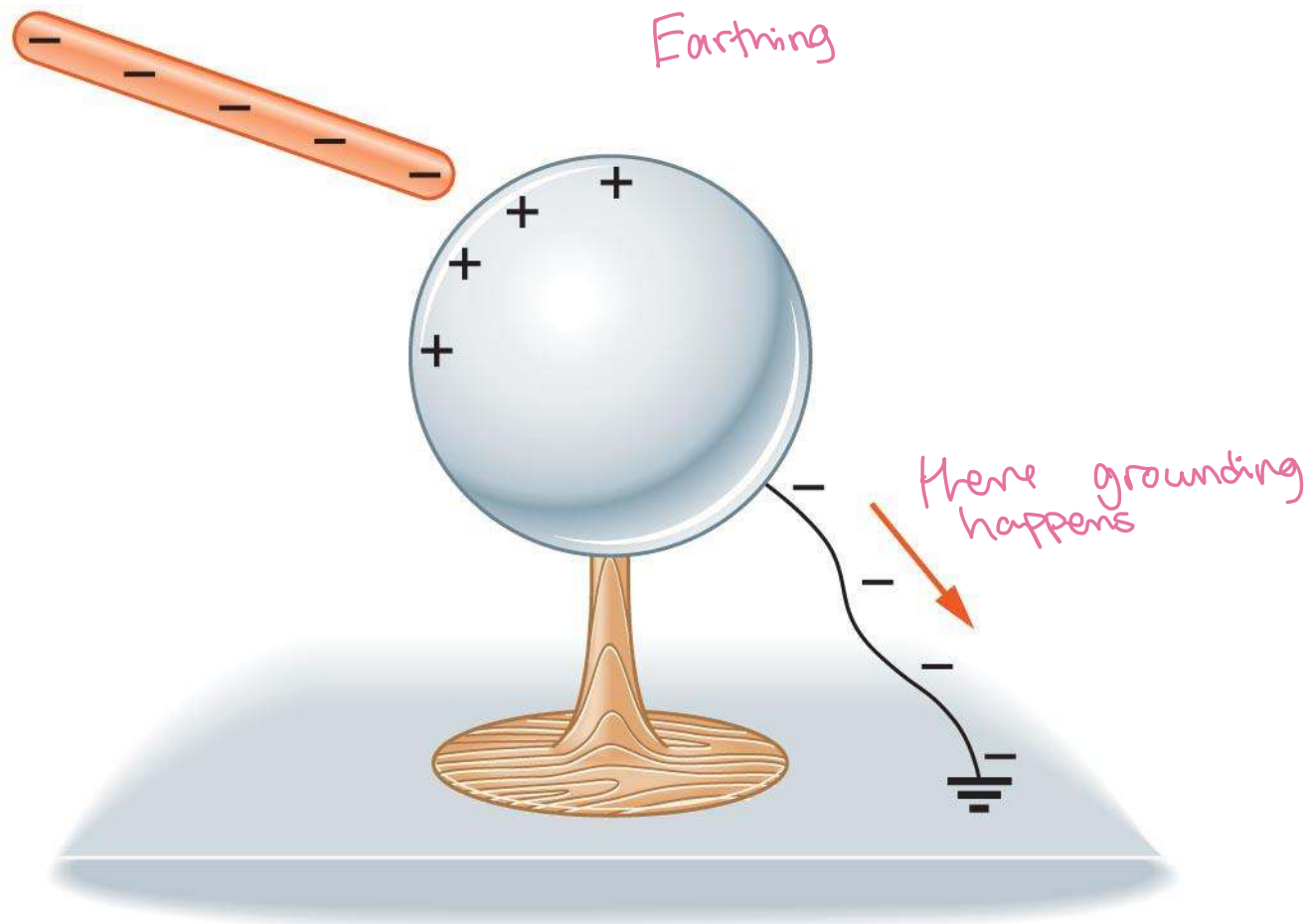
Question : Is it possible to charge a conductor by induction?

Answer: **YES**, a conductor can be charged by induction, if there is a way to ground it.

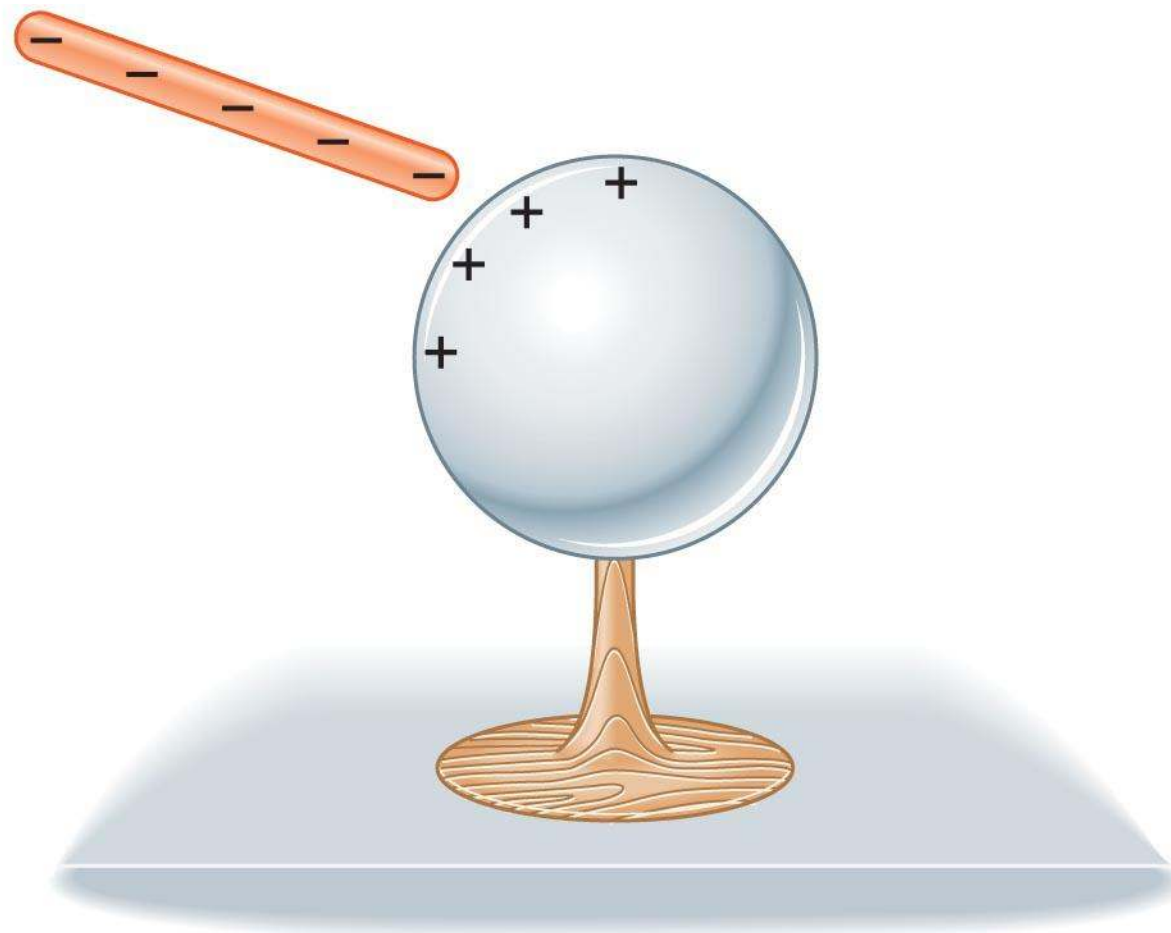
Note: When charging a conductor by induction, the final charge of the conductor will be opposite in sign to the charge of the charging rod



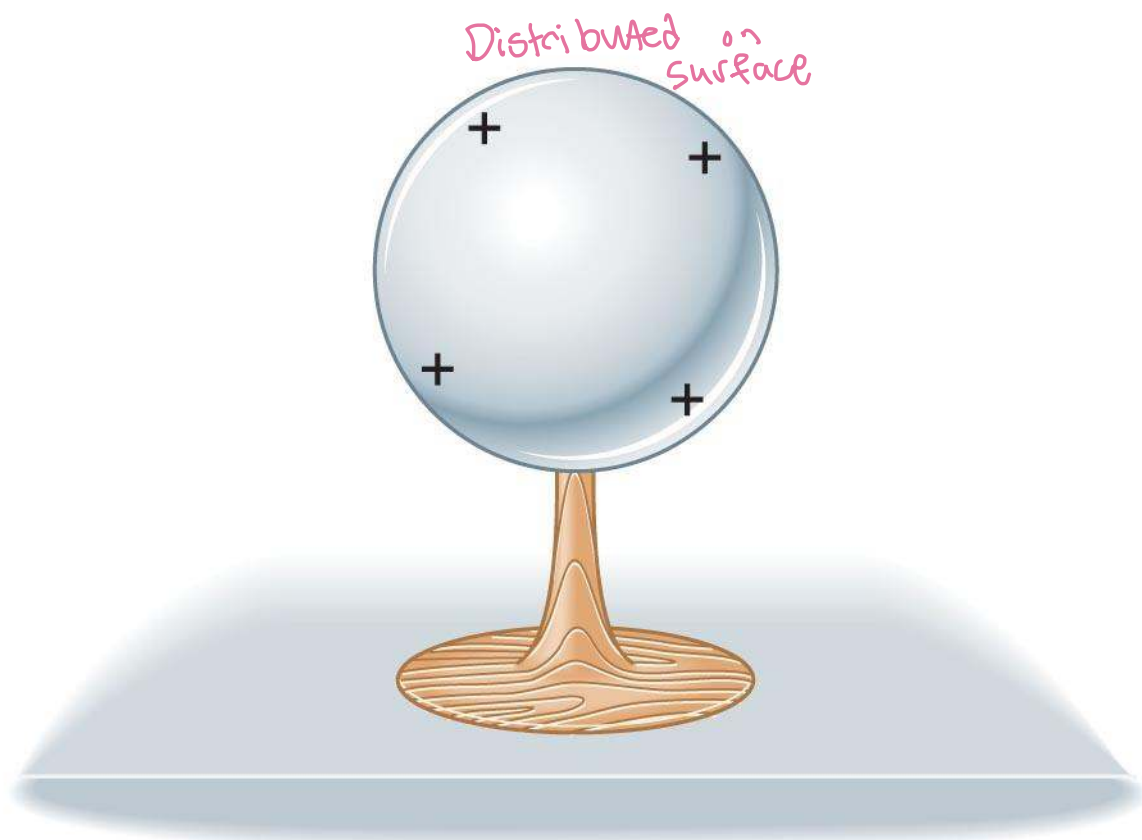
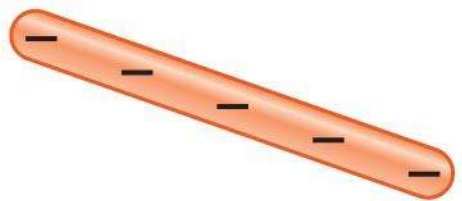
(a)



(b)



(c)



(d)

point charges are very small

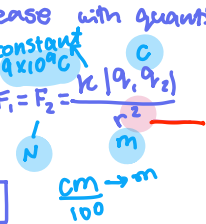


F_1 & F_2 are equal in magnitude, opposite in direction
 due to Force of q_2

q_1 is $4 \times q_2$
 what's forces? equal

repulsion will increase with quantity increase
 attraction

r : distance



$\frac{cm}{100} \rightarrow m$

Don't forget Square
 μC convert to C
 $\mu C = 10^{-6} C$ | $nC = 10^{-9} C$ | $1cm = 10^{-2} m$ | $1mm = 10^{-3} m$

Coulomb's Law

Coulomb's law gives the electrostatic force between two-point charges (**at rest**):

$$F = k \frac{|q_1||q_2|}{r^2} \quad (\text{SI unit: Newton, N})$$

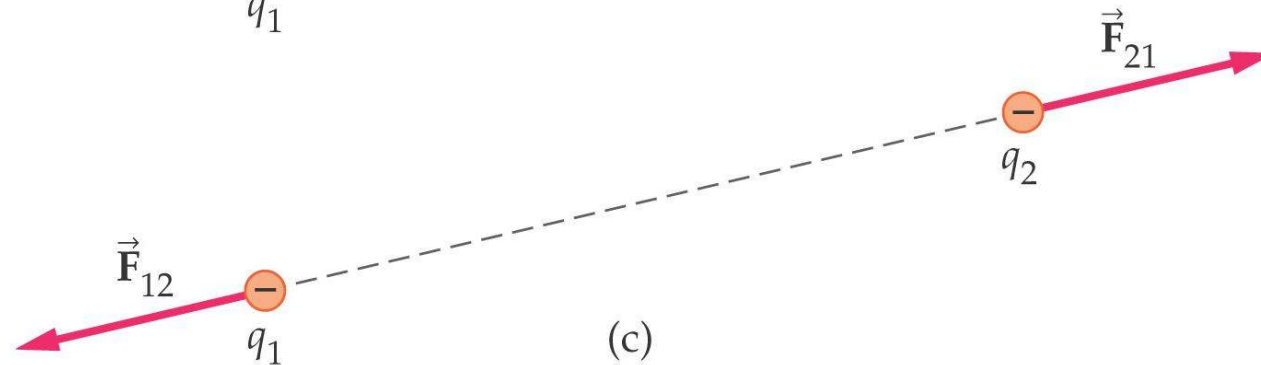
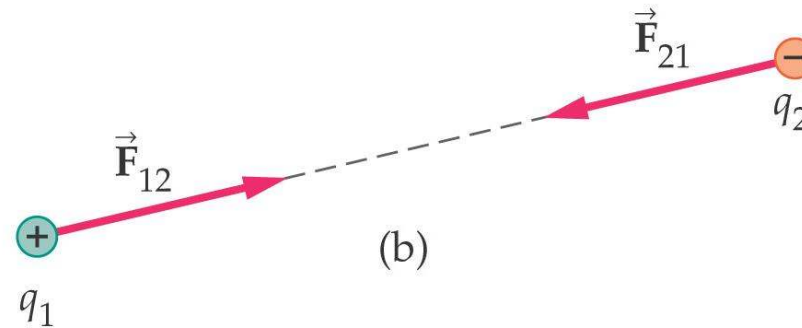
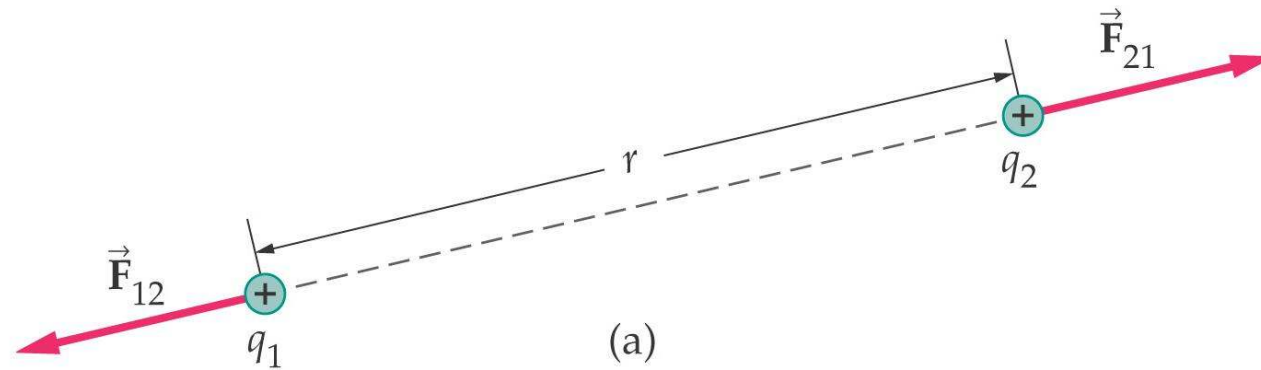
$$k = \frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2 \quad (\text{Coulomb's constant})$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2 \quad (\text{permittivity constant})$$

- ✓ **F** is along the line connecting the charges
- ✓ **F** is attractive if the charges are opposite (Fig. b)
- ✓ **F** is repulsive if the charges are alike (Figs. a and c)

The forces on the two charges are action-reaction forces

why drawing
on point?



Example 1

Two point-charges $q_1 = + 3.13 \times 10^{-6} \text{ C}$ and $q_2 = - 4.47 \times 10^{-6} \text{ C}$, are separated by a distance of 25.5 cm.

0.255 m

- 1) Find the magnitude of the electrostatic force experienced by the positive

charge $F = \frac{k |q_1 q_2|}{r^2} = \frac{9 \times 10^9 [(3.13 \times 10^{-6})(4.47 \times 10^{-6})]}{0.255^2} = 1.936 \text{ N}$

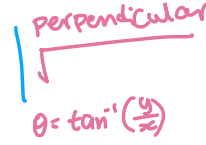
- 2) Is the magnitude of the force experienced by the negative charge greater than, less than, or the same as that experienced by the positive charge?

Explain. *same but different direction*

Multiple Forces

- When a charge experiences forces due to two or more other charges, the net force on it is simply the vector sum of the forces taken individually.
- In other words, Coulomb's law helps us to calculate the force between several charges. We calculate the forces one at a time and **ADD them AS VECTORS**.

$$\vec{F}_{1,\text{net}} = \vec{F}_{12} + \vec{F}_{13} + \vec{F}_{14} + \vec{F}_{15} + \dots + \vec{F}_{1n},$$

1D: Forces add them
2D: 
 $\theta = \tan^{-1}\left(\frac{y}{x}\right)$

Shell Theories: There are two shell theories for electrostatic force

- **Shell theory 1.** A charged particle outside a shell with charge uniformly distributed on its surface is attracted or repelled as if the shell's charge were concentrated as a particle at its center.
- **Shell theory 2.** A charged particle inside a shell with charge uniformly distributed on its surface has no net force acting on it due to the shell.

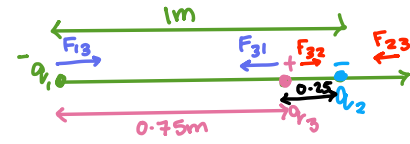
Example 2

$$\begin{aligned} q_1 &= -5.4 \mu\text{C} \rightarrow -5.4 \times 10^{-6} \text{C} \\ q_2 &= -2.2 \mu\text{C} = -2.2 \times 10^{-6} \text{C} \\ q_3 &= +1.6 \mu\text{C} = +1.6 \times 10^{-6} \text{C} \\ r &= 0.75 \text{m} \end{aligned}$$

$$F_{13} = F_{31} = \frac{k |q_1 q_3|}{r^2}$$

$$F_{13} = F_{31} = \frac{9 \times 10^9 |(-5.4 \times 10^{-6})(1.6 \times 10^{-6})|}{(0.75)^2} = 0.138 \text{ N}$$

$$F_{31} = 0.14 \text{ N Left}$$



$$F_{32} = F_{23} = \frac{k |q_2 q_3|}{r^2}$$

$$F_{32} = F_{23} = \frac{9 \times 10^9 |(2.2 \times 10^{-6})(1.6 \times 10^{-6})|}{(0.25)^2} = 0.506 \text{ N}$$

$$F_{32} = 0.506 \text{ N Right}$$

A charge $q_1 = -5.4 \mu\text{C}$ is at the origin and a charge $q_2 = -2.2 \mu\text{C}$ is set on the x axis at a distance $x = 1 \text{ m}$. Find the net force acting on a charge $q_3 = +1.6 \mu\text{C}$ located at a distance $x = 0.75 \text{ m}$.

$$\sum F_{q_3} = 0.51 - 0.14 = +0.37 \text{ N Right}$$

$$(\text{Ans. } 0.507 - 0.138 = 0.369 \text{ N})$$

Additional question

F_{net} on $q_1 = ??$ Right

$$\sum F_{q_1} = 0.14 - 0.11 = +0.03 \text{ N}$$

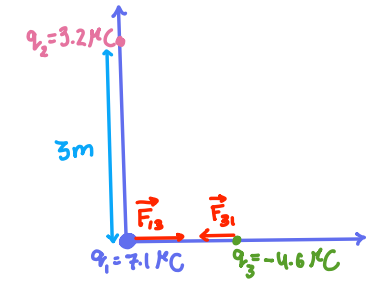
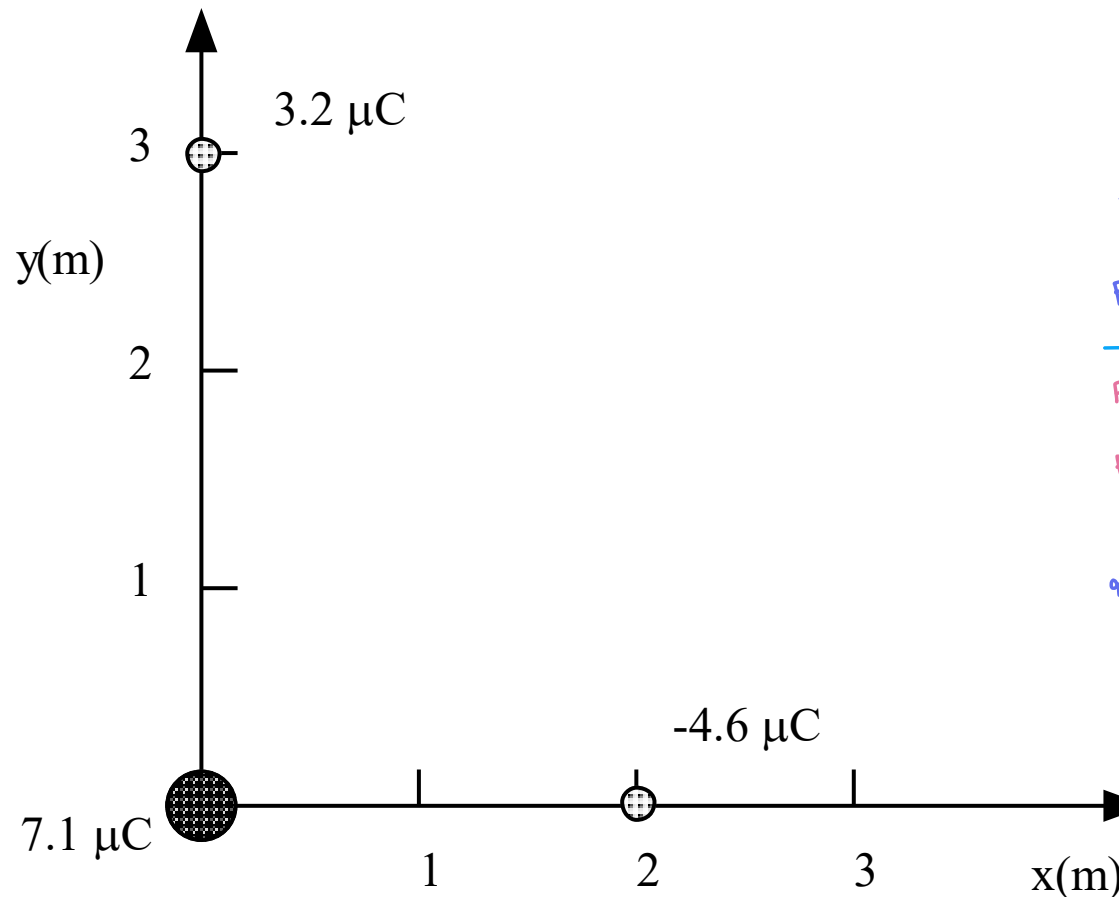
$$F_{12} = F_{21} = \frac{k |q_1 q_2|}{r^2} = \frac{9 \times 10^9 |(5.4 \times 10^{-6})(2.2 \times 10^{-6})|}{1^2} = 0.11 \text{ N Left}$$



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Example 3

In the figure below, find the magnitude and direction of the net force on the charge located at the origin



$$F_{13} = F_{31} = \frac{k|q_1 q_2|}{r^2}$$

$$F_{13} = \frac{9 \times 10^9 (7.1 \times 10^{-6})(4.6 \times 10^{-6})}{2^2} = 0.073 \text{ N Right (+x)}$$

$$F_{12} = F_{21} = \frac{k|q_1 q_2|}{r^2} =$$

$$F_{12} = \frac{9 \times 10^9 (7.1 \times 10^{-6})(3.2 \times 10^{-6})}{3^2} = 0.023 \text{ N } -\hat{y} \text{ Down}$$

$$F_{13} = 0.073 \text{ N}$$

$$F_{12} = -0.023 \text{ N}$$

$$F_{\text{net}} = \sqrt{(0.073)^2 + (-0.023)^2}$$

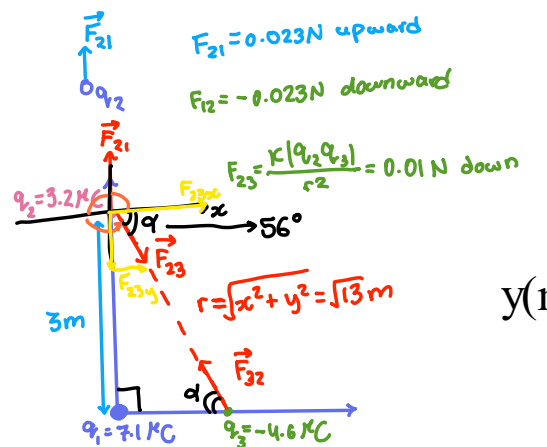
$$F_{\text{net}} = 0.076 \text{ N Down Right}$$

$$\theta = \tan^{-1}\left(\frac{F_{12}}{F_{13}}\right) = \tan^{-1}\left(\frac{-0.023}{0.073}\right) = -17.49^\circ + 360^\circ$$

$$342.5^\circ \text{ (4th quadrant)}$$

Example 4

In the figure below, find the magnitude and direction of the net force on the charge located at the $y=3\text{m}$



$$\alpha = \tan^{-1}\left(\frac{\text{opp}}{\text{adj}}\right) = \tan^{-1}\left(\frac{3}{2}\right) = 56.31^\circ \approx 56^\circ$$

*components angle no negative

$$F_{23} (0.01, 360^\circ - 56^\circ)$$

$$F_{23} (0.01, 304^\circ)$$

$$F_{23x} = F_{23} \cos 304^\circ = 0.0056 \text{ N}$$

$$F_{23y} = F_{23} \sin(304^\circ) = -0.0083 \text{ N}$$

$$F_{\text{net}x} = 0.0056 \text{ N} (+\hat{x})$$

$$F_{\text{net}y} = 0.023 - 0.0083 = 0.0147 \approx 0.015 \text{ N} (+\hat{y})$$

$$F_{\text{net}} = \sqrt{(F_{\text{net}x})^2 + (F_{\text{net}y})^2} = 0.016 \text{ N}$$

$$\theta = \tan^{-1}\left(\frac{F_{\text{net}y}}{F_{\text{net}x}}\right) = \tan^{-1}\left(\frac{0.015}{0.0056}\right) = 69.5^\circ$$

Extra practice: Find net force on q_3

$$F_{12} = F_{21} = 0.023 \text{ N}$$

$$F_{23} = F_{32} = 0.01 \text{ N}$$

$$F_{13} = F_{31} = 0.073 \text{ N}$$

$$F_{31} = -0.073 \text{ N}$$

$$F_{32x} = F_{32} \cos(124^\circ) = -0.0056 \text{ N} (-\hat{x})$$

$$F_{32y} = F_{32} \sin(124^\circ) = 0.0083 \text{ N} (+\hat{y})$$

$$F_{32} (-0.01, 124^\circ)$$

3 forces acting on q_3

① F_{32x}

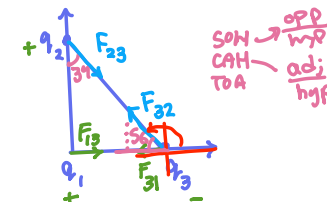
② F_{32y}

③ F_{31}

$$F_{\text{net}} = 0.079 \text{ N}$$

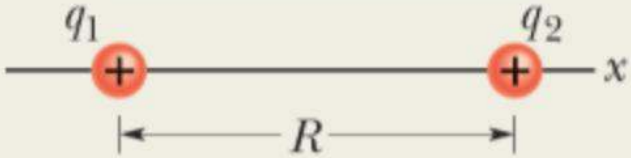
$$\theta = \tan^{-1}\left(\frac{0.0083}{-0.0786}\right) = -6.03^\circ$$

$$= 173.97^\circ$$

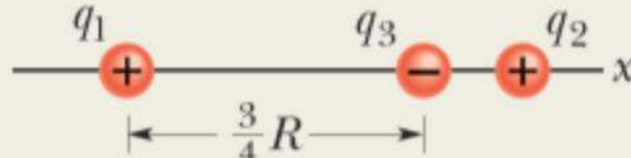


Extra Home Practice 1

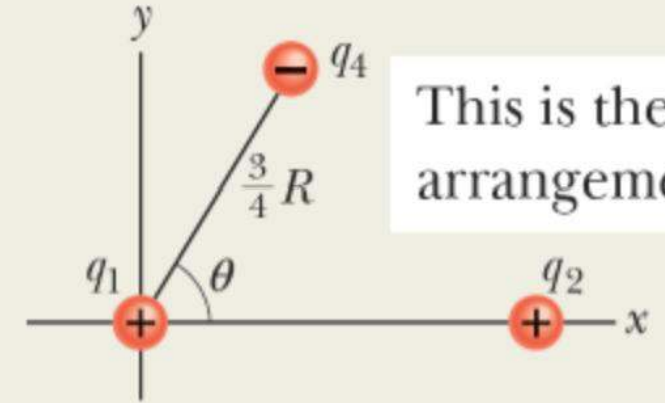
This is the first arrangement.



This is the second arrangement.



This is the third arrangement.



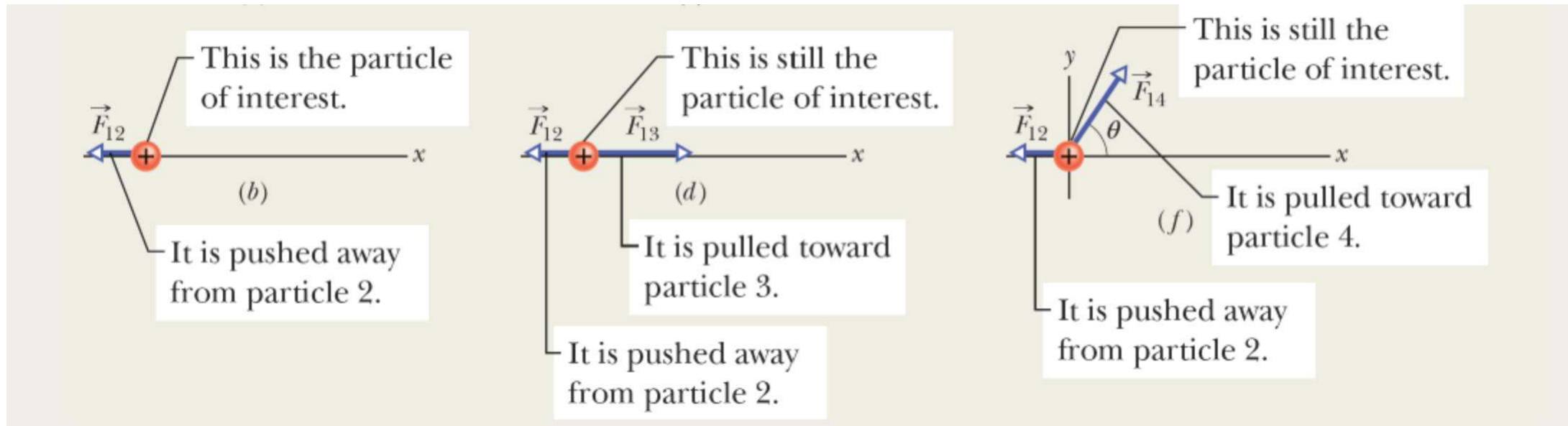
Calculate the net force on q_1 in each of the shown arrangements.

Given:

$$q_1 = 1.6 \times 10^{-19} \text{C} \quad q_2 = 3.2 \times 10^{-19} \text{C} \quad q_3 = q_4 = -3.2 \times 10^{-19} \text{C},$$

$$R = 2 \text{ cm}$$

Answer to Home Practice 1:



Answer to first arrangement

$1.15 \times 10^{-24} \text{ N}$ and 180° . (Answer)

$$\vec{F}_{12} = -(1.15 \times 10^{-24} \text{ N})\hat{i}. \quad (\text{Answer})$$

Answer to second arrangement

$$\begin{aligned} \vec{F}_{1, \text{net}} &= \vec{F}_{12} + \vec{F}_{13} \\ &= -(1.15 \times 10^{-24} \text{ N})\hat{i} + (2.05 \times 10^{-24} \text{ N})\hat{i} \\ &= (9.00 \times 10^{-25} \text{ N})\hat{i}. \end{aligned} \quad (\text{Answer})$$

Thus, $\vec{F}_{1, \text{net}}$ has the following magnitude and direction (relative to the positive direction of the x axis):

$9.00 \times 10^{-25} \text{ N}$ and 0° . (Answer)

Answer to third arrangement

$$\begin{aligned} F_{1,\text{net},x} &= F_{12,x} + F_{14,x} = F_{12} + F_{14} \cos 60^\circ \\ &= -1.15 \times 10^{-24} \text{ N} + (2.05 \times 10^{-24} \text{ N})(\cos 60^\circ) \\ &= -1.25 \times 10^{-25} \text{ N}. \end{aligned}$$

The sum of the y components gives us

$$\begin{aligned} F_{1,\text{net},y} &= F_{12,y} + F_{14,y} = 0 + F_{14} \sin 60^\circ \\ &= (2.05 \times 10^{-24} \text{ N})(\sin 60^\circ) \\ &= 1.78 \times 10^{-24} \text{ N}. \end{aligned}$$

The net force $\vec{F}_{1,\text{net}}$ has the magnitude

$$F_{1,\text{net}} = \sqrt{F_{1,\text{net},x}^2 + F_{1,\text{net},y}^2} = 1.78 \times 10^{-24} \text{ N.} \quad (\text{Answer})$$

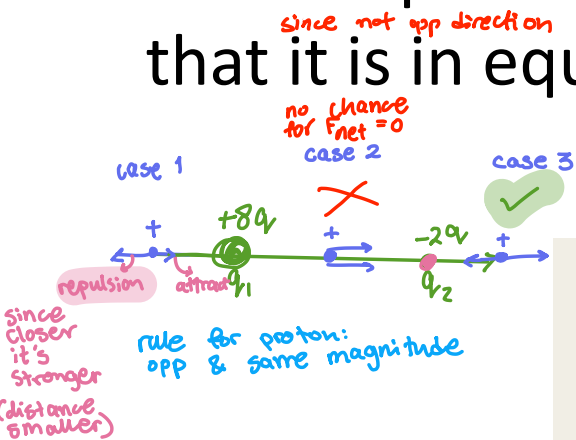
To find the direction of $\vec{F}_{1,\text{net}}$ we take

$$\theta = \tan^{-1} \frac{F_{1,\text{net},y}}{F_{1,\text{net},x}} = -86.0^\circ.$$

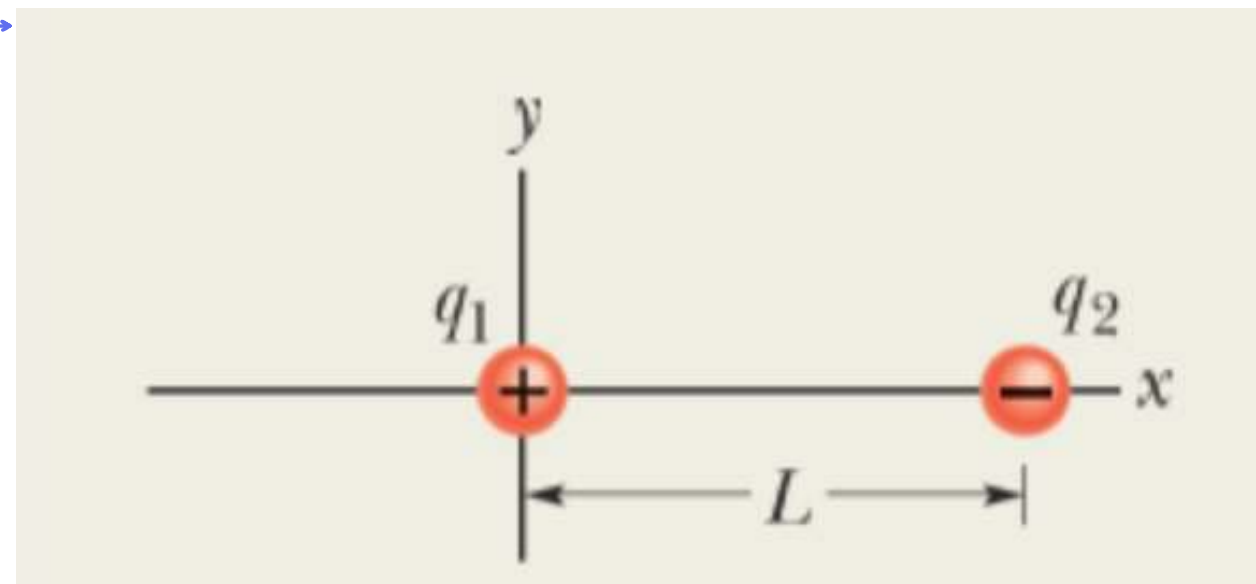
Extra Home Practice 2

cancelled
came in quiz / major
↓

At what point (other than infinitely far away) can a proton⁺ be placed so that it is in equilibrium (the net force on it is zero)? $q_1 = +8q$ and $q_2 = -2q$

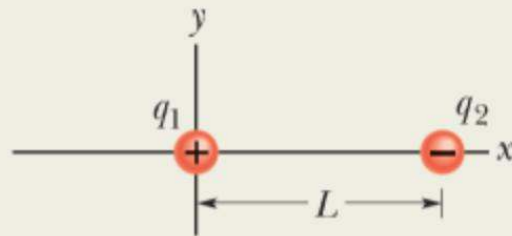


attract:
① depends on distance
②

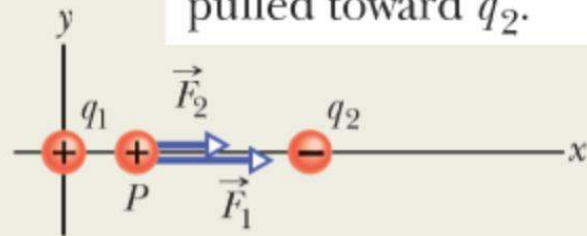


Answer to Home Practice 2:

Three possible locations P, S, and R for a proton. At each location, F_1 is the force on the proton from q_1 and F_2 is the force on the proton from q_2 .



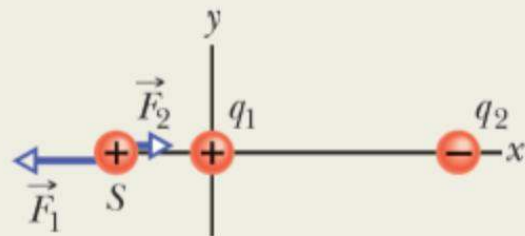
(a)



Pushed away from q_1 ,
pulled toward q_2 .

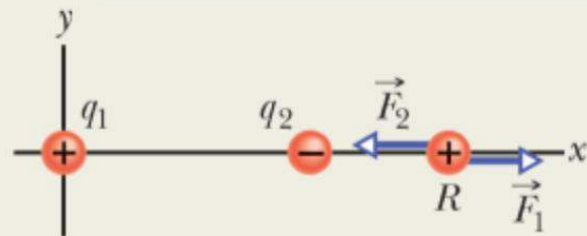
(b)

The forces cannot cancel
(same direction).



(c)

The forces cannot cancel
(one is definitely larger).



(d)

The forces can cancel,
at the right distance.

Section 21.2 Charge is Quantized

Learning Objectives can't be half electron Discrete/quantized

- Identify the elementary charge.
- Identify that the charge of a particle or object must be a positive or negative integer times the elementary charge.

Adaptive assignment → not graded

- The charge of a particle or object must be a **positive or negative integer times the elementary charge**.
- Electric charge is **quantized** (restricted to certain values). It is possible, for example, to find a particle that has no charge at all or a charge of $+10e$ or $-6e$, but not a particle with a charge of, say, $3.57e$.
- The charge of a particle can be written as ne , where n is a positive or negative integer and e is the elementary charge. Any positive or negative charge q that can be detected can be written as

$$q = ne, \quad n = \pm 1, \pm 2, \pm 3, \dots,$$

- (gain)
+ (lose)

in which **e** , the **elementary charge**, has the approximate value

$$e = 1.602 \times 10^{-19} \text{ C}.$$

Charge for Three Particles and Their Antiparticles

Particle	Symbol	Charge	Antiparticle	Symbol	Charge
Electron	e or e^-	$-e$	Positron	e^+	$+e$
Proton	p	$+e$	Antiproton	$\bar{p}$	$-e$
Neutron	n	0	Antineutron	$\bar{n}$	0

Example:

$q_1 = q_2 = q$ (identical) Modt+ Damon
↳ da vinci's code

proton mass > electron mass

Initially, sphere A has a charge of $-50e$ and sphere B has a charge of $+20e$. The spheres are made of conducting material and are **identical in size**. If the spheres then touch, what is the resulting charge on sphere A?

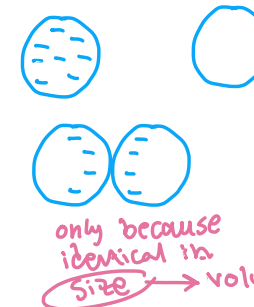
Answer: $-15e$

Random question:

$$\begin{aligned} -6e - 2e &= Q + Q \\ -8e &= \frac{2Q}{2} \\ Q &= -4e \end{aligned}$$



$$\begin{aligned} Q_{i,sys} &= Q_{f,system} \\ -50e + 20e &= Q + Q = 2Q \\ \frac{-30e}{2} &= \frac{2Q}{2} \\ Q &= -15e \end{aligned}$$



conservation of charge
 $charge_i = charge_f$

Section 21.3 Charge is Conserved

➤ The net electric charge of any isolated system is always conserved.

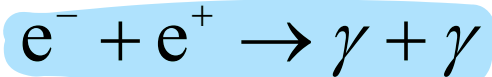
If you rub a glass rod with silk, a positive charge appears on the rod.

Measurement shows that a negative charge of equal magnitude appears on the silk. This suggests that rubbing does not create charge but only transfers it from one body to another, upsetting the electrical neutrality of each body during the process.

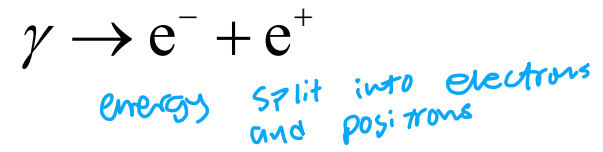
Radiation from sun
electromagnetic spectrum
photon: no mass
no charge
only packet of energy

Gamma ray most energetic photons

- If two charged particles undergo an annihilation process, they have equal and opposite signs of charge. (A **gamma ray γ** is a high-energy form of **electromagnetic radiation** (a photon) that has **no mass and no electric charge.**)



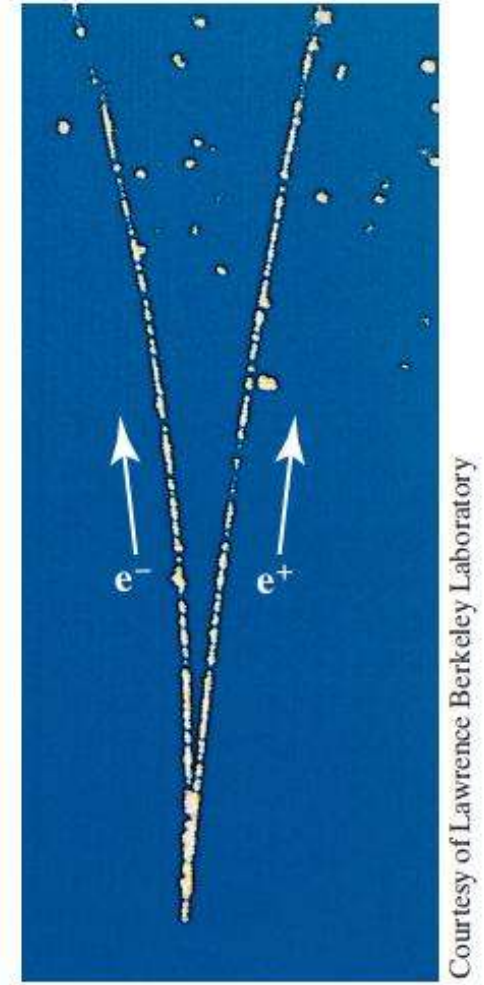
- If two charged particles appear as a result of a pair production process, they have equal and opposite signs of charge.



Experimental Evidence of Particle Charge Conservation

Real-life example

A photograph of trails of bubbles left in a bubble chamber by an electron and a positron. The pair of particles was produced by a gamma ray that entered the chamber directly from the bottom. Being electrically neutral, the gamma ray did not generate a telltale trail of bubbles along its path, as the electron and positron did.



Courtesy of Lawrence Berkeley Laboratory